Do Now · retrieval

Trees and binary search trees

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

In chain A→B→C, which node follows B?

Q2

What does LIFO mean?

Q3

What must be true before binary search on an array?

Q4

Insert X between A and B: what should new X.next point to?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Start at root — Compare the new value 25 with root 40; it belongs in the left subtree.

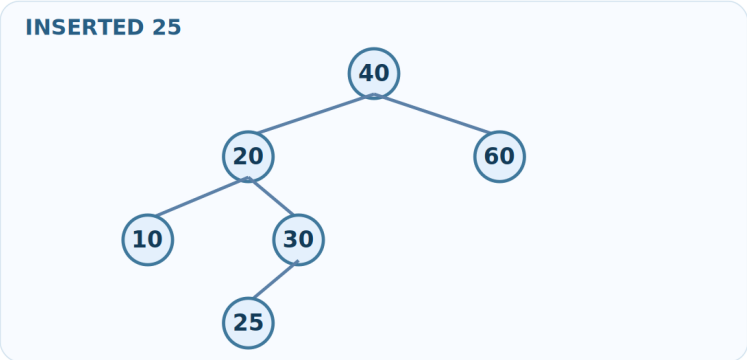
Core idea 2

Compare again — 25 is greater than 20, so move right. The right child currently stores 30.

Core idea 3

Insert without breaking the property — 25 is smaller than 30; insert it as the left child of 30.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

root — Topmost tree node.

child — Node directly below a parent.

subtree — A node and all its descendants.

binary search tree — A binary tree ordered by values, with smaller keys to the left and greater to the right.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

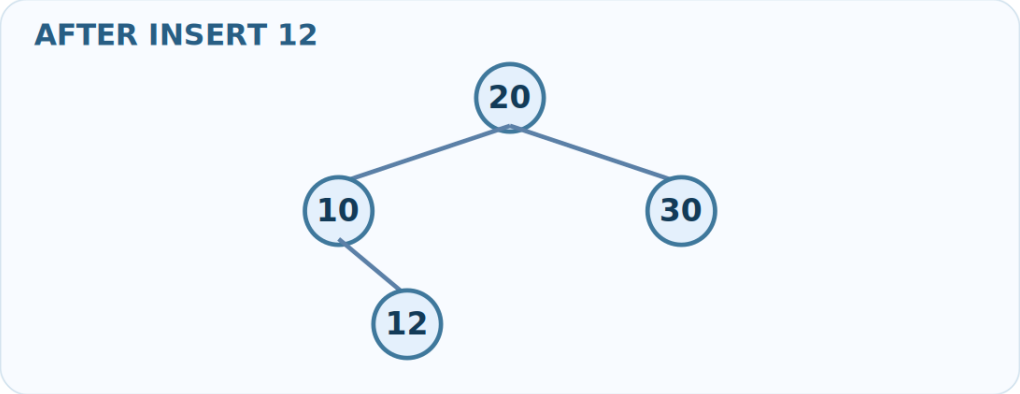
Problem

Insert 12 into BST with root 20, left 10 and right 30.

Follow the worked method

1. Predict — Compare 12 with 20 → go left to 10.
2. Confirm and explain — 12 > 10, so 12 becomes the right child of 10.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

35 is greater than 25, so go right.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

BST root 50, left child 25, right child 75: where does 60 go?

Question 2

Insert 15, 10, 20, 18 into an empty BST. Identify the parent of 18.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why a BST can have O(n) worst-case search time despite O(log n) average search on a reasonably balanced tree.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Every binary search tree has logarithmic worst-case lookup time.

Use key vocabulary from memory

1. Explain root in your own words.

2. Explain child in your own words.

3. Explain subtree in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.